

EX2:- AIM: Set up a honey pot and monitor the honey pot on the network

1. Introduction to Honeypots

A **honeypot** is a decoy system intentionally made vulnerable to entice cyber attackers. Its primary goal is to lure attackers and log their activities for later analysis. By monitoring a honeypot, network administrators can understand how attackers operate and what vulnerabilities they target. This information helps to strengthen the actual systems within an enterprise network.

2. What is PentBox?

PentBox is a security suite written in Ruby that provides tools for network analysis and penetration testing. One of its key features is setting up a simple honeypot. While it may not offer the complexity of more advanced honeypot solutions, **PentBox** is ideal for a lightweight, easily-configured honeypot.

3. Prerequisites

Before proceeding, ensure you meet the following requirements:

- **Kali Linux** installed on your system.
- Root access or appropriate privileges to install software.
- **Ruby** installed on your Kali Linux system (most versions of Kali come with Ruby pre-installed).
- A basic understanding of networking and Linux commands.

4. Step-by-Step Guide: Installing and Configuring PentBox Honeypot

4.1 Step 1: Update Your Kali Linux

To ensure that your system has the latest updates and security patches, run the following commands:

```
sudo apt update && sudo apt upgrade -y
```

4.2 Step 2: Install PentBox on Kali Linux

You can either download **PentBox** from GitHub or clone the repository directly to your system.

Run the following commands in your terminal:

```
git clone https://github.com/technicaldada/pentbox.git
```

```
cd pentbox
```

```
cd pentbox-1.8
```

4.3 Step 3: Configuring the Honeypot

To configure the honeypot, follow these steps:

1. Run PentBox by executing:

```
./pentbox.rb
```

2. Select the **Network Tools** section by typing:

2

3. From the menu, select the **Honeypot** option by typing:

3

4. You will be presented with two configuration options:

- **Fast Auto Configuration:** Automatically sets up the honeypot.
- **Manual Configuration:** Allows custom settings.

Fast Auto Configuration:

To use the Fast Auto Configuration option, type:

1

Next, select a **port** to monitor (e.g., port 80 for HTTP or port 22 for SSH). After selecting the port, PentBox will notify you that the **HONEYPOT ACTIVATED ON PORT 80** (or whichever port you selected).

Manual Configuration:

For Manual Configuration, follow these steps:

- Select **Manual Configuration** by typing:

2

- Set up the port number you want the honeypot to monitor (e.g., 23 for Telnet) and enter a custom warning message to show to attackers, like:
- You are not allowed to access my system!
- Save the log file by confirming the default path:
- `*/pentbox/other/log_honeypot.txt`
- After completing the configuration, you will see a notification that the **HONEYPOT ACTIVATED** on your chosen port.

4.4 Step 4: Running the Honeypot

Once the honeypot is configured, it will begin listening on the selected port. For example, if you configured the honeypot to listen on port 80 (HTTP), any unauthorized attempts to access your system will trigger alerts.

4.5 Step 5: Monitoring the Honeypot

As attackers interact with your honeypot, the terminal will display messages such as:

```
INTRUSION ATTEMPT DETECTED from <IP>:<Port>
```

This allows you to monitor incoming attacks and observe any suspicious activity.

5. Verifying Honeypot Activity

You can manually test the honeypot by opening a new terminal and running a simple test. For example, open your web browser, type the honeypot's IP address in the address bar (e.g., `http://10.0.2.15`), and press Enter. If the honeypot is configured correctly, you will receive an "Access Denied" message or the custom warning you defined during setup.

6. Analyzing Logs

PentBox logs all detected intrusion attempts in a text file. You can analyze these logs later to see details like the IP address of the attacker, the port number used, and the nature of the intrusion attempt. The log file is saved at the default location:

```
*/pentbox/other/log_honeypot.txt
```

To view the logs, use the following command:

```
cat /pentbox/other/log_honeypot.txt
```

7. Conclusion

Setting up a honeypot with **PentBox** on **Kali Linux** is a simple yet effective way to detect unauthorized network access attempts. Though **PentBox** is not the most sophisticated honeypot tool available, it provides a quick and easy way to monitor and log intrusion attempts. This helps you gain insight into how attackers might target your systems, allowing you to improve your security posture.

Advanced Topics: Expanding and Enhancing Your Honeypot Setup

Once you have set up a basic honeypot with **PentBox**, you might want to expand its capabilities or enhance the monitoring to make it more effective for specific use cases. Below are some advanced topics you can explore to improve your honeypot deployment:

8. Honeypot Enhancement Techniques

8.1 Simulating Different Services

You can configure your honeypot to simulate various network services that are commonly targeted by attackers. By doing so, you can increase the chances of attracting intruders and gaining more insight into different types of attacks.

For example, if you want to simulate a vulnerable **FTP** or **SMTP** server, you can choose to configure the honeypot to listen on the corresponding ports (21 for FTP, 25 for SMTP). Attackers who are probing for open services will likely attempt to exploit these services, giving you a chance to analyze their methods.

8.2 Creating Multiple Honeypots on Different Ports

You don't have to limit your honeypot to a single service or port. You can run **multiple honeypots** on different ports simultaneously to increase the likelihood of catching attackers. Here's how you can do it:

- Set up PentBox to listen on several common ports:
 - **Port 22** (SSH)
 - **Port 23** (Telnet)
 - **Port 80** (HTTP)
 - **Port 443** (HTTPS)

For each service, you can configure custom warning messages and logging for intrusions. Running multiple honeypots at the same time diversifies the potential attack surfaces, attracting different types of attack vectors.

8.3 Running PentBox on a Dedicated Machine

For a more realistic honeypot deployment, consider setting up **PentBox** on a dedicated machine separate from your production network. By doing this, you avoid interference with your actual systems and give attackers a more believable target. Ensure that this dedicated system is isolated so that any successful attacks on the honeypot do not lead to deeper network breaches.

8.4 Logging and Alerts

While PentBox logs attack details to a file by default, you can improve monitoring by integrating real-time alerting mechanisms. Some suggestions include:

- **Email Alerts:** Configure a script that monitors the log file and sends an email alert when a new intrusion attempt is detected.
- **Syslog Integration:** Forward PentBox logs to a central logging server using Syslog. This allows you to aggregate and analyze logs from multiple sources.
- **Integration with SIEM:** If your organization uses a **Security Information and Event Management** (SIEM) system, you can integrate PentBox logs into the SIEM for real-time analysis, correlation, and alerting.

9. Analyzing Honeypot Data

The logs generated by the honeypot contain valuable information that can help you understand the methods and tools used by attackers. Here are a few ways to analyze and extract meaningful insights from the collected data.

9.1 Log Parsing and Attack Profiling

PentBox logs each intrusion attempt with details such as the attacker's IP address, the port number, and sometimes the content of the packet. By manually parsing these logs, you can:

- **Identify attack patterns:** If certain IPs or regions repeatedly attempt to exploit the honeypot, you may be dealing with a persistent threat.
- **Recognize specific vulnerabilities:** If attackers frequently target specific ports or services, this may indicate which vulnerabilities are being widely exploited in the wild.
- **Profile attacker behavior:** You can analyze how attackers interact with the honeypot to understand their motives, techniques, and tools.

9.2 Visualizing Attack Data

You can use data visualization tools to map out attack patterns. Export the honeypot logs to tools like **ELK Stack** (Elasticsearch, Logstash, Kibana), **Splunk**, or **Grafana** to create dashboards that:

- Show geographic locations of attackers.
- Track the frequency of attacks over time.
- Identify which services (ports) are most frequently targeted.

9.3 Identifying the Source of Attacks

To gain additional context about the attacks, you can cross-reference IP addresses from your honeypot logs with external threat intelligence sources. Tools like **Shodan** or public IP blacklists can help you identify whether the attacking IPs are part of known malicious networks.

10. Improving Honeypot Realism

To make your honeypot more believable and attractive to attackers, you can take the following steps:

10.1 Mimic Real Services

Ensure that the services running on your honeypot look realistic. For example:

- Set up a fake **SSH server** with fake login credentials.
- Simulate a **web server** with a vulnerable version of a popular application (e.g., an outdated WordPress installation).
- Emulate a **database server** that appears to store sensitive information.

10.2 Adding Fake Content

Populate your honeypot with files, directories, or databases that contain fake data. Attackers will be more likely to stay and explore if they believe they've found something of value. Examples of fake content include:

- A directory named **"/sensitive_files"** with decoy documents.
- Fake login credentials or database tables that appear to store personal data.

These tactics help keep the attackers engaged, providing you with more data about their behavior.

11. Further Securing Your Honeypot Environment

Although honeypots are designed to be compromised, you don't want attackers to use them as a stepping stone into your real network. Here are a few strategies to ensure that your honeypot remains isolated and secure:

11.1 Network Isolation

Place your honeypot in a **demilitarized zone (DMZ)** or on a **separate VLAN** to ensure that, even if the honeypot is compromised, the attacker cannot gain access to your production systems. You can also configure firewalls to limit incoming traffic to specific services you want to monitor.

11.2 Limit Outbound Communication

Restrict the honeypot's ability to initiate outbound communication. This prevents attackers from using your honeypot as a **launch point** to carry out attacks against other systems on the internet or within your network.

11.3 Regularly Update Honeypot Software

While the honeypot itself is intended to be vulnerable, keeping the underlying system (such as Kali Linux or PentBox) up to date ensures that attackers cannot exploit the system hosting the honeypot. Use tools like **Unattended Upgrades** to automate the process of applying security patches.

12. Legal and Ethical Considerations

Before deploying a honeypot, be aware of the legal and ethical implications. Depending on your jurisdiction, the following points should be considered:

- **Informed Consent:** Ensure that deploying a honeypot in an enterprise network is compliant with company policies and legal guidelines. Unauthorized monitoring could violate privacy laws.
- **Data Retention:** Follow legal regulations regarding how long you can store data about attackers, especially if their personal information can be identified.
- **Avoid Counterattacks:** While monitoring attacker behavior is legitimate, launching counterattacks or actively engaging in retaliation could lead to legal consequences.

Always consult with legal counsel and ensure you are following appropriate cybersecurity best practices and regulations when using honeypots.

Conclusion

By following this extended guide, you can not only set up a basic honeypot using PentBox but also expand its functionality to create a more robust, effective, and secure honeypot environment. Whether you are using it to monitor general attack traffic or as part of a broader defense strategy, a properly configured honeypot can serve as a powerful tool in understanding and mitigating potential threats to your network.